

Q1 (3)

 E_1 : Event denotes spade is missing

$$P(E_1) = \frac{1}{4}; P(\overline{E_1}) = \frac{3}{4}$$

A: Event drawn two cards are spade

$$P(A) = \frac{\frac{1}{4} \times \left(\frac{{}^{12}C_2}{{}^{51}C_2} \right) + \frac{3}{4} \times \left(\frac{{}^{13}C_2}{{}^{51}C_2} \right) + \frac{3}{4} \times \left(\frac{{}^{13}C_2}{{}^{51}C_2} \right)}{\frac{1}{4} \times \left(\frac{{}^{12}C_2}{{}^{51}C_2} \right) + \frac{3}{4} \times \left(\frac{{}^{13}C_2}{{}^{51}C_2} \right)}$$

$$= \frac{39}{50}$$

Q2 (2)

Total cases :

$$\underline{6} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2}$$

$$n(s) = 6 \cdot 6!$$

Favourable cases :

Number divisible by 3 $\equiv$

Sum of digits must be divisible by 3

Case-I

1,2,3,4,5,6

Number of ways = 6!

Case-II

0,1,2,4,5,6

Number of ways = 5 · 5!

Case-III

0,1,2,3,4,5

Number of ways = 5 · 5!

$$n(\text{favourable}) = 6! + 2 \cdot 5 \cdot 5!$$

$$P = \frac{6! + 2 \cdot 5 \cdot 5!}{6 \cdot 6!} = \frac{4}{9}$$

Hints and Solutions

MathonGo

Q3 (2)

$$n(E) = 5 + 4 + 4 + 3 + 1 = 17$$

$$\text{So, } P(E) = \frac{17}{36}$$

Q4 (6)

$$\text{Let } P(E_1) = P_1; P(E_2) = P_2; P(E_3) = P_3$$

$$P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3) = \alpha = P_1(1 - P_2)(1 - P_3) \dots \dots (1)$$

$$P(\bar{E}_1 \cap E_2 \cap \bar{E}_3) = \beta = (1 - P_1)P_2(1 - P_3) \dots \dots (2)$$

$$P(\bar{E}_1 \cap \bar{E}_2 \cap E_3) = \gamma = (1 - P_1)(1 - P_2)P_3 \dots \dots (3)$$

$$P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3) = P = (1 - P_1)(1 - P_2)(1 - P_3) \dots \dots (4)$$

$$\text{Given that, } (\alpha - 2\beta)P = \alpha\beta$$

$$(P_1(1 - P_2)(1 - P_3) - 2(1 - P_1)P_2(1 - P_3))P = P_1P_2$$

$$(1 - P_1)(1 - P_2)(1 - P_3)^2$$

$$\Rightarrow (P_1(1 - P_2) - 2(1 - P_1)P_2) = P_1P_2$$

$$\Rightarrow (P_1 - P_1P_2 - 2P_2 + 2P_1P_2) = P_1P_2$$

$$\Rightarrow P_1 = 2P_2 \dots \dots (1)$$

$$\text{and similarly, } (\beta - 3\gamma)P = 2B\gamma$$

$$P_2 = 3P_3 \dots \dots (2)$$

$$\text{So, } P_1 = 6P_3 \Rightarrow \frac{P_1}{P_3} = 6$$

Q5 (4)

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ \text{odd place} & \text{even place} & \text{odd place} & \text{even place} \end{array}$$

or

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ \text{even place} & \text{odd place} & \text{even place} & \text{odd place} \end{array}$$

$$\Rightarrow \left(\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{3} \right) + \left(\frac{2}{2} \cdot \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} \right)$$

$$\Rightarrow \frac{1}{9}$$

Hints and Solutions

MathonGo

Q6 (1)

$$P(X = 1) = {}^5C_1 \cdot p \cdot q^4 = 0.4096$$

$$P(X = 2) = {}^5C_2 \cdot p^2 \cdot q^3 = 0.2048$$

$$\Rightarrow \frac{q}{2p} = 2$$

$$\Rightarrow q = 4p \text{ and } p + q = 1$$

$$\Rightarrow p = \frac{1}{5} \text{ and } q = \frac{4}{5}$$

Now

$$P(X = 3) = {}^5C_3 \cdot \left(\frac{1}{5}\right)^3 \cdot \left(\frac{4}{5}\right)^2 = \frac{10 \times 16}{125 \times 25} = \frac{32}{625}$$

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

A pack of cards has one card missing. Two cards are drawn randomly and are found to be spades. The probability that the missing card is not a spade, is :

- (1) $\frac{3}{4}$
- (2) $\frac{52}{867}$
- (3) $\frac{39}{50}$
- (4) $\frac{22}{425}$

Q2: 16 March (Shift 2) - Single Correct

Let A denote the event that a 6 -digit integer formed by 0,1,2,3,4,5,6 without repetitions, be divisible by 3. Then probability of event A is equal to :

- (1) $\frac{9}{56}$
- (2) $\frac{4}{9}$
- (3) $\frac{3}{7}$
- (4) $\frac{11}{27}$

Q3: 17 March (Shift 1) - Single Correct

Two dices are rolled. If both dices have six faces numbered 1,2,3,5,7 and 11, then the probability that the sum of the numbers on the top faces is less than or equal to 8 is :

- (1) $\frac{4}{9}$
- (2) $\frac{17}{36}$
- (3) $\frac{5}{12}$
- (4) $\frac{1}{2}$

Q4: 17 March (Shift 1) - Numerical

Let there be three independent events E_1 , E_2 and E_3 . The probability that only E_1 occurs is α , only E_2 occurs is β and only E_3 occurs is γ . Let 'p' denote the probability of none of events occurs that satisfies the equations

Questions with Answer Keys

MathonGo

$(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval $(0,1)$

Then, $\frac{\text{Pr obability of occurrence of } E_1}{\text{Probability of occurrence of } E_3}$ is equal to _____

Q5: 17 March (Shift 2) - Single Correct

Let a computer program generate only the digits 0 and 1 to form a string of binary numbers with probability of occurrence of 0 at even places be $\frac{1}{2}$ and probability of occurrence of 0 at the odd place be $\frac{1}{3}$. Then the probability that '10' is followed by '01' is equal to :

(1) $\frac{1}{18}$

(2) $\frac{1}{3}$

(3) $\frac{1}{6}$

(4) $\frac{1}{9}$

Q6: 18 March (Shift 2) - Single Correct

Let in a Binomial distribution, consisting of 5 independent trials, probabilities of exactly 1 and 2 successes be 0.4096 and 0.2048 respectively. Then the probability of getting exactly 3 successes is equal to:

(1) $\frac{32}{625}$

(2) $\frac{80}{243}$

(3) $\frac{40}{243}$

(4) $\frac{128}{625}$

Answer Key**Q1 (3)****Q2 (2)****Q3 (2)****Q4 (6)****Q5 (4)****Q6 (1)**